



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

MASTER THESIS PROPOSAL

**Unsupervised 3D Scene Reconstruction from Sparse
LiDAR**

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Contents

1	Introduction	3
2	Background	4
2.1	LiDAR-Based Semantic Scene Completion	4
2.2	Unsupervised and Self-Supervised 3D Learning	5
2.3	Neural Scene Representations for Outdoor Environments	6
2.4	Geometric Distance Metrics for 3D Reconstruction	7
3	Research Gap and Problem Statement	9
4	Proposed Methodology	10
4.1	Framework Overview	10
4.2	Voxel Encoder–Decoder Network	10
4.3	Temporal Aggregation for Pseudo-Dense Supervision	11
4.4	Unsupervised Loss Formulation	11
5	Experimental Design	12
5.1	Dataset and Benchmark	12
5.2	Evaluation Metrics	13
5.3	Baselines and Comparisons	13
6	Expected Contributions	13
7	Preliminary Timeline	14

1 Introduction

The rapid advancement of autonomous driving and robotic navigation has placed three-dimensional scene understanding at the forefront of modern perception research. LiDAR (Light Detection and Ranging) sensors have become a cornerstone of outdoor 3D perception pipelines, providing precise geometric measurements of the surrounding environment [1, 2]. However, due to physical constraints such as limited angular resolution and beam count, raw LiDAR scans produce inherently sparse point clouds that capture only a fraction of the true scene geometry. This sparsity is further exacerbated at longer ranges and in the presence of occlusions caused by dynamic objects such as vehicles and pedestrians [3].

To bridge the gap between sparse sensor observations and the dense 3D representations required for safe navigation, the task of Semantic Scene Completion (SSC) has emerged as a critical research direction. SSC aims to jointly predict the occupancy and semantic class of every voxel in a defined 3D volume from partial observations [4]. While supervised approaches have achieved notable performance on established benchmarks [5, 6, 7], they are fundamentally dependent on dense, voxel-level ground-truth annotations. The acquisition of such annotations is prohibitively expensive and labor-intensive, requiring specialized equipment and extensive manual correction. This dependency on labeled data constitutes a significant bottleneck that limits the scalability and generalizability of current methods to new environments and sensor configurations.

Unsupervised and self-supervised learning paradigms offer a principled alternative by leveraging geometric priors and structural regularities inherent in outdoor environments, such as ground planarity, surface continuity, and temporal consistency across sequential scans [8, 9]. Despite their promise, the application of unsupervised methodologies to full-scale outdoor 3D scene completion remains underexplored. Existing unsupervised efforts have predominantly focused on object-level completion using synthetic datasets [10, 11], and their extension to large-scale, real-world driving scenarios with complex occlusion patterns and dynamic content presents substantial open challenges.

Consequently, the primary objective of this thesis is to investigate and develop an unsupervised learning framework for 3D scene reconstruction from sparse LiDAR point clouds. The proposed approach formulates the task as voxel-based occupancy prediction, leveraging geometric consistency losses and structural priors to train without ground-truth dense annotations. The framework will be evaluated on the SemanticKITTI Scene Completion benchmark [1, 12], enabling direct comparison with established supervised base-

lines and providing a quantitative assessment of the viability of unsupervised methods for this task.

2 Background

To establish a comprehensive foundation for this proposal, it is essential to examine the current state of research across three interconnected domains. This section outlines existing methodologies, recent advancements, and the critical challenges that motivate an unsupervised approach to 3D scene reconstruction from sparse LiDAR data.

2.1 LiDAR-Based Semantic Scene Completion

The task of Semantic Scene Completion (SSC) was formally introduced by Song et al. [4], who proposed SSCNet to jointly infer the occupancy and semantic labels of a voxelized 3D volume from a single depth image. This formulation established the standard pipeline: given a partial observation of a scene, the model must predict the complete volumetric representation including regions that are unobserved or occluded. The task has since become a central benchmark for evaluating holistic 3D scene understanding in autonomous driving contexts [3, 12].

Subsequent research has extended SSCNet along multiple axes. LMSCNet [6] introduced a lightweight multi-scale architecture that significantly reduced computational cost while maintaining competitive accuracy, making real-time deployment more feasible. S3CNet [13] addressed the challenge of sparse LiDAR inputs directly, designing a network that operates on raw point clouds rather than dense voxelized inputs. More recently, VoxFormer [5] applied the Transformer architecture to sparse voxel queries, achieving state-of-the-art results on the SemanticKITTI benchmark by decoupling geometry completion from semantic labeling. Camera-based approaches such as MonoScene [7] have also demonstrated that monocular images alone can produce reasonable volumetric predictions, though with reduced geometric precision compared to LiDAR-based methods.

Multi-modal fusion has emerged as another active direction. Lu et al. [14] proposed a continuous voxelized fusion strategy that integrates LiDAR and camera features, while Xiao et al. [15] introduced instance-level awareness into monocular SSC to improve the completion of individual objects. Sze and Kunze [16] focused on real-time efficiency through memory-efficient sparse convolution, addressing the deployment constraints of autonomous vehicles. Yang et al. [17] proposed a semantic segmentation-assisted ap-

proach that leverages 2D semantic predictions to guide 3D scene completion, demonstrating the benefit of auxiliary semantic cues for improving geometric reconstruction. Rist et al. [18] explored deep implicit functions as an alternative to discrete voxel grids, enabling continuous geometric representations from LiDAR data.

Despite this progress, a fundamental limitation persists across all the above methods: they rely on dense voxel-level ground-truth annotations for training. The SemanticKITTI Scene Completion benchmark [1] provides such annotations by aggregating multiple LiDAR scans and projecting semantic labels into a voxel grid, but this process is sensor-specific, environment-specific, and does not scale to new domains without significant re-annotation effort. This dependency on supervised labels motivates the exploration of unsupervised alternatives.

2.2 Unsupervised and Self-Supervised 3D Learning

The prohibitive cost of obtaining dense 3D annotations has driven significant research interest toward learning paradigms that reduce or eliminate the need for ground-truth labels. This section examines three related but distinct threads: self-supervised depth completion, unsupervised point cloud completion, and self-supervised occupancy prediction.

Self-supervised depth completion methods aim to produce dense depth maps from sparse LiDAR inputs combined with monocular camera images. Ma et al. [9] introduced a foundational framework that trains a sparse-to-dense network using photometric consistency between the predicted dense depth and the observed camera image, requiring no ground-truth dense depth maps. Song et al. [19] extended this paradigm by integrating visual-LiDAR odometry to enforce geometric consistency across consecutive frames. For extremely sparse inputs, Hu et al. [20] demonstrated self-supervised depth completion from a single-line LiDAR, pushing the boundary of how little geometric supervision is required. Feng et al. [21] further advanced self-supervised monocular depth estimation by using sparse LiDAR measurements as a complementary training signal rather than as direct supervision. While these methods successfully densify depth representations, they operate in 2.5D (depth image space) rather than producing full volumetric 3D reconstructions, limiting their applicability to downstream tasks that require complete scene geometry.

In the domain of unsupervised point cloud completion, recent work has focused on learning to reconstruct complete 3D shapes without access to ground-truth complete point clouds. Wu et al. [10] proposed a multi-view adversarial learning framework that leverages rendered 2D projections from

multiple viewpoints as a supervisory signal, enabling completion through cross-view geometric agreement. Lee et al. [11] formulated the completion task as an unbalanced optimal transport problem, providing a principled mathematical framework for matching partial observations to complete shapes without paired training data. However, both approaches have been evaluated primarily on synthetic object-level datasets such as ShapeNet, and their scalability to large-scale outdoor driving environments with hundreds of dynamic objects and complex occlusion patterns remains undemonstrated.

A more recent and directly relevant thread is self-supervised occupancy prediction. SelfOcc [22] demonstrated that 3D occupancy representations can be learned from camera inputs alone using neural rendering as the supervisory signal, bypassing the need for voxel-level annotations entirely. OccFeat [23] approached the problem from a pretraining perspective, learning occupancy features in a self-supervised manner to improve downstream Bird’s Eye View segmentation. Occupancy-MAE [24] applied masked autoencoder pretraining to large-scale LiDAR point clouds, learning occupancy representations by reconstructing randomly masked voxel regions. Lange et al. [25] extended self-supervised occupancy learning to the temporal dimension, predicting multiple future occupancy states without ground-truth annotations. Vizzo et al. [8] proposed a self-supervised scan completion method specifically designed for sparse outdoor LiDAR, exploiting temporal aggregation of sequential scans as a proxy for ground-truth dense geometry. While these approaches demonstrate the feasibility of reducing label dependency, they either focus on pretraining representations for supervised fine-tuning, or address partial aspects of the completion task rather than full unsupervised scene reconstruction from sparse LiDAR inputs.

2.3 Neural Scene Representations for Outdoor Environments

An alternative paradigm for 3D scene reconstruction has emerged through neural scene representations, which learn continuous implicit or explicit functions to model geometry and appearance. These methods are inherently self-supervised in their training procedure, as they optimize scene representations through differentiable rendering losses without requiring 3D ground-truth annotations.

Neural Radiance Fields (NeRF) have been adapted to outdoor driving contexts by several recent works. Lightning NeRF [26] introduced a hybrid representation combining explicit voxel grids with implicit neural features to accelerate both training and rendering for large-scale driving scenes. S-

NeRF++ [27] extended the NeRF framework to support autonomous driving simulation, enabling both reconstruction of existing scenes and generation of novel traffic configurations. DriveEnv-NeRF [28] explored the use of NeRF-based reconstructions as virtual environments for validating autonomous driving systems, demonstrating the practical utility of high-fidelity neural scene models. Shen et al. [29] provided a broader survey of neural rendering techniques applied to urban scene reconstruction, identifying both the potential and the current limitations of these approaches for autonomous driving applications.

Despite their capacity to produce visually compelling reconstructions, neural scene representations face several fundamental limitations in the context of this thesis. First, most NeRF-based methods require per-scene optimization, meaning a separate model must be trained for each individual scene rather than learning a generalizable reconstruction function across scenes. This makes them impractical for real-time deployment or large-scale evaluation across diverse environments. Second, these methods are primarily designed for static scenes and struggle with dynamic objects such as moving vehicles and pedestrians, which are prevalent in driving scenarios. Third, their output is a continuous radiance or density field rather than a discrete voxel occupancy grid, making direct comparison against established SSC benchmarks such as SemanticKITTI non-trivial without additional post-processing steps. For these reasons, neural scene representations, while promising for visualization and simulation tasks, are not directly suited to the structured evaluation framework of semantic scene completion benchmarks, and their application to unsupervised volumetric reconstruction from sparse LiDAR remains an open research direction.

2.4 Geometric Distance Metrics for 3D Reconstruction

Unsupervised 3D reconstruction methods depend critically on the choice of geometric loss function, as no ground-truth dense geometry is available during training. The loss must quantify the discrepancy between a predicted 3D structure and an observed partial measurement, and its properties directly influence training stability, convergence behavior, and the quality of reconstructed outputs. This section reviews the principal distance metrics used in point cloud and voxel-based reconstruction, with emphasis on recent variants that address known failure modes of the standard formulation.

The Chamfer Distance (CD) is the most widely adopted metric for evaluating and training point cloud reconstruction models. Given a predicted

point set P and a reference point set Q , the symmetric Chamfer Distance is defined as:

$$\text{CD}(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} \|p - q\|_2^2 + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} \|q - p\|_2^2 \quad (1)$$

The first term penalizes predicted points that lack nearby counterparts in the reference (accuracy), while the second penalizes reference points not covered by the prediction (completeness). Despite its simplicity and differentiability, the standard CD suffers from well-documented pathological behaviors. Wu et al. [30] demonstrated that CD is insensitive to local density variations: a prediction that clusters points densely in some regions while leaving others uncovered can achieve a low CD score, producing visually implausible reconstructions. To address this, they proposed Density-aware Chamfer Distance (DCD), which incorporates a density-penalizing term that encourages uniform coverage of the target surface.

Several subsequent works have proposed alternative modifications to the CD formulation. Lin et al. [31] introduced Hyperbolic Chamfer Distance, which embeds point representations in hyperbolic space to better capture hierarchical geometric structures, achieving improved completion quality on objects with complex topology. The same research group later proposed InfoCD [32], a contrastive formulation that reinterprets the completion loss through the lens of mutual information maximization, providing stronger gradients for training and reducing sensitivity to outliers. Huang et al. [33] proposed Learnable Chamfer Distance, which parameterizes the distance function itself as a trainable module, allowing the loss to adapt to the specific geometric characteristics of the reconstruction task during training. Lu et al. [34] developed Oriented and Directional Chamfer Distance variants that incorporate surface normal information into the distance computation, improving reconstruction accuracy for objects with thin structures and sharp edges where standard Euclidean nearest-neighbor matching is ambiguous.

In the voxelized setting relevant to semantic scene completion, the reconstruction objective is typically formulated as per-voxel binary classification (occupied vs. free) rather than point-to-point distance minimization. Binary cross-entropy and its class-balanced variants are the standard losses for supervised SSC. For unsupervised training, however, the loss must be derived from geometric consistency between predicted voxel occupancy and observed point clouds or multi-frame aggregations, making the choice and adaptation of geometric distance metrics a central design decision in the proposed framework.

3 Research Gap and Problem Statement

The preceding literature review reveals a clear gap at the intersection of three research threads. Supervised semantic scene completion methods (Section 2.1) achieve strong performance but are fundamentally constrained by the requirement for dense voxel-level annotations that are expensive to acquire and do not transfer across sensor configurations or environments. Unsupervised and self-supervised 3D learning approaches (Section 2.2) have demonstrated that meaningful geometric representations can be learned without dense labels, but existing work has either focused on 2.5D depth completion, object-level point cloud completion on synthetic data, or pretraining representations intended for subsequent supervised fine-tuning. Neural scene representations (Section 2.3) offer an alternative reconstruction paradigm but require per-scene optimization and produce continuous fields that are incompatible with the discrete voxel evaluation framework of established SSC benchmarks. Furthermore, the geometric distance metrics reviewed in Section 2.4 exhibit pathological behaviors under standard formulations, adding complexity to the design of unsupervised training objectives that must operate without direct supervision.

To date, no existing method has demonstrated fully unsupervised scene completion from sparse LiDAR point clouds at the scale and evaluation rigor of standard benchmarks such as SemanticKITTI. The specific open challenges that this thesis aims to address are:

1. **Scalability of unsupervised completion to full outdoor scenes.** Existing unsupervised point cloud completion methods [10, 11] operate on isolated synthetic objects. Extending these to large-scale outdoor environments with hundreds of objects, complex occlusion patterns, and significant depth variation is non-trivial.
2. **Geometric supervision without dense ground truth.** The unsupervised training signal must be derived entirely from the sparse observations themselves. Temporal aggregation of sequential LiDAR scans [8] provides a proxy for denser geometry, but designing loss functions that effectively exploit this signal for voxel-level occupancy prediction requires careful formulation.
3. **Evaluation against established supervised baselines.** The viability of unsupervised methods must be assessed quantitatively on the same benchmarks used by supervised approaches. Understanding the performance gap and identifying where unsupervised methods succeed

or fail relative to their supervised counterparts is essential for guiding future research.

Formally, given a sparse LiDAR point cloud $\mathcal{X}_t \subset \mathbb{R}^3$ captured at time t , and a sequence of temporally adjacent scans $\{\mathcal{X}_{t-k}, \dots, \mathcal{X}_{t+k}\}$ with known ego-motion transformations, the objective is to learn a function $f_\theta : \mathcal{X}_t \rightarrow \hat{\mathcal{V}} \in \{0, 1\}^{H \times W \times D}$ that predicts the binary occupancy of each voxel in a predefined 3D volume, *without* access to ground-truth voxel labels $\mathcal{V}^*$ during training.

4 Proposed Methodology

This section presents the proposed unsupervised framework for 3D scene reconstruction from sparse LiDAR. The method consists of three principal components: a voxel-based encoder–decoder network for occupancy prediction, a temporal aggregation module that constructs pseudo-dense supervision from sequential scans, and a set of geometric consistency losses that provide the unsupervised training signal.

4.1 Framework Overview

The overall pipeline operates as follows. At training time, a sparse LiDAR scan $\mathcal{X}_t$ is voxelized into a binary input tensor $\mathcal{V}_{\text{in}} \in \{0, 1\}^{H \times W \times D}$, where each voxel is marked as occupied if it contains at least one LiDAR return point. This sparse input is processed by a 3D encoder–decoder network that produces a dense occupancy probability map $\hat{\mathcal{V}} \in [0, 1]^{H \times W \times D}$. Concurrently, the temporal aggregation module transforms and merges neighboring scans $\{\mathcal{X}_{t-k}, \dots, \mathcal{X}_{t+k}\}$ into the coordinate frame of $\mathcal{X}_t$ using ego-motion estimates, producing a pseudo-dense reference volume $\mathcal{V}_{\text{agg}}$. The network parameters θ are optimized by minimizing a composite loss that measures geometric consistency between $\hat{\mathcal{V}}$ and $\mathcal{V}_{\text{agg}}$, supplemented by structural regularization priors. At inference time, only the single-scan input path is used; the temporal aggregation module is not required.

4.2 Voxel Encoder–Decoder Network

The encoder–decoder architecture follows the established U-Net paradigm adapted for 3D sparse convolution, drawing on design principles from LM-SCNet [6] and S3CNet [13]. The encoder applies a series of sparse 3D convolutional blocks with residual connections and progressive downsampling to

extract multi-scale geometric features from the sparse input volume. Skip connections transmit high-resolution features to the decoder, which applies transposed sparse convolutions to upsample the representation back to the original voxel resolution. The final output layer applies a sigmoid activation to produce per-voxel occupancy probabilities.

Sparse convolution is essential for computational efficiency, as the vast majority of voxels in outdoor LiDAR volumes are unoccupied. By operating only on non-empty voxels and their neighborhoods, the network avoids the cubic memory and computation costs of dense 3D convolution, following the efficiency principles demonstrated by Sze and Kunze [16].

4.3 Temporal Aggregation for Pseudo-Dense Supervision

The core insight enabling unsupervised training is that while any individual LiDAR scan is sparse, the aggregation of multiple temporally adjacent scans observed from different ego-vehicle positions yields a substantially denser representation of the static scene geometry [8]. This aggregated volume serves as a pseudo-ground-truth supervisory signal.

Given a sequence of scans $\{\mathcal{X}_{t-k}, \dots, \mathcal{X}_{t+k}\}$ and the corresponding ego-motion transformations $\{T_{t-k \rightarrow t}, \dots, T_{t+k \rightarrow t}\}$ (obtained from LiDAR odometry or GPS/IMU), each neighboring scan is transformed into the coordinate frame of the reference scan $\mathcal{X}_t$ and voxelized. The union of all transformed voxel occupancies produces $\mathcal{V}_{\text{agg}}$. To mitigate noise introduced by dynamic objects that move between frames, a statistical filtering step removes voxels with inconsistent occupancy across the temporal window: voxels occupied in fewer than a threshold fraction of contributing scans are discarded. This approach follows the principle established by Vizzo et al. [8] and is analogous to the multi-scan aggregation used to construct the SemanticKITTI ground truth [1], but applied automatically without manual annotation.

4.4 Unsupervised Loss Formulation

The training objective combines three complementary loss terms:

Occupancy consistency loss. The primary loss measures the agreement between predicted occupancy $\hat{\mathcal{V}}$ and the pseudo-dense reference $\mathcal{V}_{\text{agg}}$ using binary cross-entropy applied only to voxels where $\mathcal{V}_{\text{agg}}$ provides reliable information (i.e., voxels that are either consistently occupied or confirmed free

by ray-casting through the aggregated scans):

$$\mathcal{L}_{\text{occ}} = -\frac{1}{|\mathcal{M}|} \sum_{v \in \mathcal{M}} \left[\mathcal{V}_{\text{agg}}(v) \log \hat{\mathcal{V}}(v) + (1 - \mathcal{V}_{\text{agg}}(v)) \log(1 - \hat{\mathcal{V}}(v)) \right] \quad (2)$$

where $\mathcal{M}$ denotes the set of voxels with reliable pseudo-labels.

Geometric smoothness prior. A total-variation regularizer encourages spatial coherence in the predicted occupancy, penalizing isolated spurious voxels and encouraging smooth surfaces:

$$\mathcal{L}_{\text{smooth}} = \sum_v \sum_{v' \in \mathcal{N}(v)} |\hat{\mathcal{V}}(v) - \hat{\mathcal{V}}(v')| \quad (3)$$

where $\mathcal{N}(v)$ denotes the 6-connected spatial neighborhood of voxel v .

Ground plane constraint. Outdoor driving scenes exhibit a strong structural prior: the ground surface is approximately planar and should be consistently occupied in the lowest voxel layers. A constraint term encourages high occupancy probability for voxels at the estimated ground height, derived from a RANSAC plane fit to the lowest LiDAR returns:

$$\mathcal{L}_{\text{ground}} = -\frac{1}{|\mathcal{G}|} \sum_{v \in \mathcal{G}} \log \hat{\mathcal{V}}(v) \quad (4)$$

where $\mathcal{G}$ is the set of voxels on the estimated ground plane.

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{occ}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_g \mathcal{L}_{\text{ground}} \quad (5)$$

where λ_s and λ_g are weighting hyperparameters to be determined through ablation.

5 Experimental Design

5.1 Dataset and Benchmark

The proposed method will be evaluated on the SemanticKITTI Scene Completion benchmark [1, 12], which provides voxelized ground-truth occupancy and semantic labels for 22 driving sequences captured by a Velodyne HDL-64E LiDAR sensor. The standard split uses sequences 00–10 for training (with sequence 08 reserved for validation) and sequences 11–21 for testing.

The completion volume is defined as $256 \times 256 \times 32$ voxels at 0.2 m resolution, covering $51.2 \text{ m} \times 51.2 \text{ m} \times 6.4 \text{ m}$ in front of the ego-vehicle. Additionally, the Occ3D benchmark [35] will be considered as a secondary evaluation to assess generalizability across different annotation pipelines and sensor configurations.

5.2 Evaluation Metrics

Following standard SSC evaluation protocol, the primary metric for geometry completion (the focus of this thesis) is the Intersection over Union (IoU) computed over all occupied voxels regardless of semantic class:

$$\text{IoU} = \frac{|\hat{\mathcal{V}}_{\text{occ}} \cap \mathcal{V}_{\text{occ}}^*|}{|\hat{\mathcal{V}}_{\text{occ}} \cup \mathcal{V}_{\text{occ}}^*|} \quad (6)$$

where $\hat{\mathcal{V}}_{\text{occ}}$ and $\mathcal{V}_{\text{occ}}^*$ denote the sets of voxels predicted and labeled as occupied, respectively. Precision and recall will also be reported to distinguish between accuracy and completeness of the reconstruction. The analysis will additionally examine performance stratified by distance from the ego-vehicle, as reconstruction difficulty increases with range due to increasing point cloud sparsity.

5.3 Baselines and Comparisons

The experimental comparison will include the following categories:

- **Supervised SSC methods** (upper-bound reference): LMSCNet [6], S3CNet [13], VoxFormer [5], and MonoScene [7].
- **Self-supervised methods**: Vizzo et al. [8] (self-supervised scan completion), Occupancy-MAE [24] (self-supervised pretraining), and Self-Occ [22] (self-supervised occupancy from cameras).
- **Ablated variants of the proposed method**: to isolate the contribution of each loss component ($\mathcal{L}_{\text{occ}}$, $\mathcal{L}_{\text{smooth}}$, $\mathcal{L}_{\text{ground}}$), the temporal aggregation window size k , and the dynamic object filtering threshold.

6 Expected Contributions

This thesis is expected to make the following contributions:

1. An unsupervised learning framework for voxel-based 3D scene reconstruction from sparse LiDAR that requires no dense ground-truth annotations, combining temporal aggregation with geometric consistency losses and structural priors.
2. A systematic quantitative comparison between the proposed unsupervised approach and established supervised baselines on the SemanticKITTI Scene Completion benchmark, providing empirical evidence on the viability and current limitations of label-free methods for this task.
3. An analysis of the conditions under which unsupervised reconstruction succeeds or fails (e.g., distance ranges, scene complexity, dynamic object density), informing future research directions in self-supervised 3D perception.

7 Preliminary Timeline

Period	Activities
Months 1–2	Literature review finalization; dataset preparation and preprocessing pipeline (SemanticKITTI voxelization, ego-motion extraction, temporal aggregation implementation).
Months 3–4	Implementation of the voxel encoder–decoder network; baseline reproduction (LMSCNet supervised training) for validation of the evaluation pipeline.
Months 5–6	Implementation of the unsupervised training pipeline (pseudo-dense supervision generation, loss formulation); initial training experiments and hyperparameter tuning (λ_s , λ_g , k).
Months 7–8	Comprehensive evaluation on SemanticKITTI; ablation studies; distance-stratified analysis; comparison against baselines. Secondary evaluation on Occ3D if time permits.
Months 9–10	Results analysis; thesis writing; revision and submission.

Table 1: Preliminary research timeline.

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